Jahnavi Yelamanchi

+1 609-240-3318 | j.yelamanchi@nyu.edu | LinkedIn | GitHub | Website

Summary

ML graduate student at NYU specializing in robotics and hardware acceleration, with prior R&D experience at NUS and IIT-B. Focused on world models, multimodal perception, and RL for safe & generalizable robots.

Education

New York University — Aug. 2024 – Present
Master's in Computer Engineering — *Advisor: Lerrel Pinto*

Vellore Institute of Technology — Sept. 2020 – June 2024
Bachelor's in Computer Science Engineering — *GPA: 3.97/4.0*

- Visiting student at the National University of Singapore, Dec. 2022 – May 2023

Experience

NYU GRAIL (Generalizable Robotics and AI Lab) — April 2025 – Present
Graduate Researcher — *Mentor: Prof. Lerrel Pinto*

- Developing tactile-centric world models using the AnySkin tactile sensor for reorientation tasks in LIBERO.
- Built a custom "push-to-align" task (rigid block + cabinet) with dynamics and latent space modeling for reinforcement learning.
- Presented initial results (Aug 2025) on DreamerV2-style multimodal world model integrating tactile + visual input

System & Artificial Intelligence (SAI) Lab — Jan 2025 – April 2025
Graduate Researcher — *Mentor: Prof. Sai Zhang*

- Accelerated Efficient-VAR model inference by integrating OpenAI Triton for kernel-level optimization, reducing latency by 17%.
- Designed a non-invasive instrumentation framework to visualize tradeoffs across top-k, top-p, and classifier-free guidance settings without altering model logic.

National University of Singapore (NUS) — 2022 – 2023
Deep Learning Research Intern — *Mentor: Prof. Liu Lili*

- Achieved 93% accuracy in multilingual language recognition using advanced neural network architectures.
- Processed 11,000+ diverse text samples with custom NLP pipelines to enable successful model training.
- Deployed models on Microsoft Azure, improving scalability and throughput.

IIT Bombay (E-Yantra Robotics Competition) — 2022
Software Development Team Lead

- Distinguished among top 25 out of 500+ participants by clearing Stage 1 of E-Yantra robotics competition.
- Implemented a Linear Quadratic Regulator (LQR) controller in CoppeliaSim for system identification, improving control accuracy by 20%.

Publications

Prediction of Cerebrospinal Fluid (CSF) Pressure with Generative Adversarial Network Synthetic Plasma-CSF Biomarker Pairing
Phani Paladugu, Rahul Kumar, Jahnavi Yelamanchi, et al.
Neuroinformatics, Vol. 23, Issue 3, p. 38. Springer US, 2025.

Performance Evaluation of Generative Adversarial Networks for Anime Face Synthesis Using Deep Learning Approaches
Perepi Rajarajeswari, Redgrowthu Vijaya Saraswathi, Jahnavi Yelamanchi, et al.
Multidisciplinary Science Journal, Vol. 7, Issue 3, Article 2025093, 2025.

Artificial Intelligence-Based Methodologies for Early Diagnostic Precision and Personalized Therapeutic Strategies in Neuro-Ophthalmic and Neurodegenerative Pathologies
Rahul Kumar, Ethan Waisberg, Jahnavi Yelamanchi, et al.
Brain Sciences, Vol. 14, Issue 12, Article 1266. MDPI, 2024.

Projects

Weight Prediction with Tactile World Models | *PyTorch, Reskin Tactile Sensors* 2025

- Built a tactile-centric world model to predict object weight using force feedback on an xArm + scale setup.

xArm Cup Picking Task | *PyTorch, LIBERO, Reskin Tactile Sensors* 2025

- Collected and processed real-world tactile + proprioceptive data for xArm cup-picking demos, enabling evaluation of multimodal policy robustness.

Curiosity-driven Robot Reinforcement Learning | *PyTorch, PPO, Random Network Distillation* 2024

- Integrated curiosity-based intrinsic motivation into a PPO robot policy using Random Network Distillation (RND), observing reduced performance in robotic tasks compared to RND's success in video games.

Trajectory Optimization and Control for Quadrotors | *Python, SQP, MPC* 2024

- Designed SQP-based trajectory controllers and MPC policies for quadrotor tracking under thrust constraints, improving accuracy by 30–40%.

Quadrotor Obstacle Avoidance with RL | *C++, PyTorch, PPO, Xcode Instruments* 2024

- Developed a custom RL environment and PPO controller with Stable-Baselines3 for quadrotor obstacle avoidance, improving path efficiency by 20%.

Teaching

CSE1006 Blockchain and Cryptocurrency Technologies Vellore Institute of Technology
Instructor: Jaisankar N, Winter 2023

CSE4001 Parallel and Distributed Computing Vellore Institute of Technology
Instructor: Nalini N, Fall 2023

Technical Skills

Programming Languages: Python, C/C++, MATLAB, Bash
Libraries & Frameworks: PyTorch, JAX, TensorFlow, Stable-Baselines3, OpenCV, Open3D, Robosuite, ROS, Mujoco
Software & Tools: Git, Docker